Garvit Mehra

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	7.32
Dehli Public School, Faridabad	Class XII (CBSE)	2024	92.60%
Dehli Public School, Faridabad	Class X (CBSE)	2022	96.60%

EXPERIENCE

Technical Intern | Xebia

May 2025 - June 2025

- Studied **LLM architectures, Transformers, RAG, fine-tuning, and memory-efficient inference across numerical precisions**.
- Explored **agentic AI, tool usage, memory, multi-agent systems, and MCP** for integrating external tools and data.
- Applied these concepts to initiate **Mosaic**, an MCP-integrated multi-agent AI platform.

POSITIONS OF RESPONSIBILITY

Technical Core Member -- Machine Learning Domain | Tinkerers' Lab, IIT Hyderabad

- Collaborated on **ML/deep learning initiatives**, contributing to model development, experimentation, debugging, and technical discussions.
- Independently developed and delivered a **1-hour hands-on reinforcement learning workshop for 150+ students**, covering RL fundamentals and **Gym LunarLander**.

PROJECTS

YOLO-Based Real-Time Object Detection System

GitHub Repo

- Built a Python-based object detection application using YOLO for inference across images, recorded videos, and live webcam streams.
- Integrated OpenCV for frame acquisition, preprocessing, annotation, and real-time visualization of detection results.
- Designed a unified inference pipeline supporting both offline and real-time input sources.
- Evaluated practical trade-offs between detection accuracy and inference speed and documented implementation learnings in a published technical article.

Mosaic -- Modular Multi-Agent AI Platform

Active Development | *GitHub Repo*

- Architected a **multi-agent AI platform** using **LangGraph ReAct agents** and an **LLM-based router** coordinating specialized web, RAG, and dynamically provisioned MCP agents.
- Built a **provider-agnostic AI architecture** supporting configurable LLM backends, **runtime MCP discovery, dynamic tool registration**, transport fallback, and hot-reloading of external services without application restarts.
- Developed **persistent multi-user conversation management** and **per-user MCP isolation** using PostgreSQL, with stateless request handling and **real-time token streaming** through Server-Sent Events.
- Engineered production infrastructure with **Redis-backed distributed rate limiting, authentication, Docker, health monitoring, and structured diagnostics**; achieved **~989 authenticated req/s at 11 ms P95 latency with 0% errors** in API load testing.

Atlas -- Enterprise Retrieval-Augmented Generation Platform

GitHub Repo

- Built a modular **RAG pipeline** using **FAISS, BM25, and mxbai-embed-large**, combining semantic and keyword

retrieval for enterprise knowledge search.
- Implemented **document chunking, hybrid retrieval, and configurable LLM inference**, keeping retrieval and generation components independently replaceable.

ACHIEVEMENTS

Motive AI Hackathon -- 9th Place

- Placed **9th** in a two-member team for developing an **enterprise knowledge RAG system**, competing against undergraduate, master's, and doctoral participants and earning eligibility for an **expedited Motive interview process**.

RELEVANT COURSES

DeepLearning.AI: Machine Learning, Advanced Learning Algorithms, Deep Learning, CNNs, Hyperparameter Tuning & Optimization

SKILLS

ML & Deep Learning: Regression, Classification, Ensemble Methods, CNNs, RNNs/LSTMs, Transformers, Transfer Learning

Frameworks: TensorFlow, Keras, PyTorch, Scikit-learn, OpenCV

NLP & GenAI: Embeddings, Semantic/Hybrid Retrieval, BM25, LLMs, RAG, Fine-Tuning, LoRA, ReAct Agents, LangGraph, LangChain, MCP, FAISS

TECHNICAL WRITING

- Published a **Medium article on building a YOLO-based real-time object detection system**. | Read Article